Behavioral and Explanation-Based Monitoring of Image Classifier Degradation Under Progressive Corruption

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Abstract*—Deep neural networks can achieve high accuracy under clean test conditions while exhibiting substantial behavioral changes when their inputs are corrupted. Detecting such degradation is important because a decline in reliability may not be fully captured by a single performance metric. This study investigates whether changes in model confidence, predictive distributions, prediction decisions, classification margins, and visual explanations can jointly characterize degradation in an image classifier under progressively increasing image corruption. A ResNet-50 classifier trained on CIFAR-10 is evaluated on the CIFAR-10-C benchmark, comprising 15 corruption types and five severity levels, resulting in 75 corruption–severity conditions. Five degradation signals are extracted: confidence change, Jensen–Shannon divergence, prediction flip rate, prediction-margin degradation, and Grad-CAM explanation instability. These signals are standardized and combined with equal weights to construct a Behavioral Degradation Index (BDI). The BDI is evaluated using descriptive correlation and monotonicity analyses, an adaptive multi-level degradation framework, Q95 warning thresholds, leave-one-corruption-out validation, multicollinearity analysis, multiple regression, and leave-one-signal-out ablation. The BDI exhibits a positive association with corruption severity, with Spearman $\rho = 0.5239$, while Grad-CAM instability provides the strongest individual severity association ($\rho = 0.5387$). However, the proposed warning mechanism does not produce advance warnings of the defined performance-failure threshold under the primary Q95 protocol, and the same behavior persists under leave-one-corruption-out calibration. The results therefore support the use of behavioral and explanation signals for degradation characterization, while indicating that substantially different calibration or modeling strategies are required for reliable early-warning detection.

***Index Terms*—deep learning, model degradation, image corruption, CIFAR-10-C, explainable artificial intelligence, Grad-CAM, behavioral monitoring, failure detection, robustness, early warning**

I. Introduction

Deep neural networks have achieved high performance across a wide range of image classification and medical imaging tasks, motivating their deployment in increasingly consequential decision-support systems. However, high accuracy on clean benchmark data does not guarantee reliable behavior when the input distribution changes. Image corruptions, distribution shifts, and other perturbations can alter both the predictive performance and the internal decision behavior of a model. In practical deployment, this creates an important monitoring problem: a model may continue to produce predictions while its confidence, decisions, and evidence patterns gradually change. Such changes can be difficult to identify using accuracy alone, particularly when ground-truth labels are unavailable at inference time. Previous work has consequently investigated failure detection, uncertainty estimation, robustness to corrupted inputs, and model degradation from several complementary perspectives [1]–[4].

Confidence-based failure detection provides one possible mechanism for identifying unreliable predictions. Existing studies have examined confidence scores, uncertainty estimation, and failure-detection performance under both in-distribution and shifted conditions. Bernhardt *et al.* investigated failure detection in medical image classification and showed that commonly used confidence measures can provide useful information about prediction reliability, while also highlighting limitations in their generalization [1]. Jaeger *et al.* further emphasized that failure-detection performance can depend strongly on the distribution shift and the evaluation protocol [2]. Similarly, confidence-based prediction-failure estimation has demonstrated that learned confidence can serve as an indicator of incorrect predictions across several image

classification and segmentation datasets [3]. These findings suggest that confidence is useful but insufficient as a universal indicator of model degradation.

A separate body of research has focused on robustness against common image corruptions. PRIME demonstrated that carefully designed augmentation primitives can improve robustness to common corruptions across datasets such as CIFAR-10-C, CIFAR-100-C, and ImageNet-C [5]. More recently, MedMNIST-C examined realistic corruption processes in medical imaging and showed that domain-specific corruption simulation can improve robustness [6]. Progressive stress-testing studies have also demonstrated that robustness varies according to architecture, perturbation type, severity, and dataset characteristics [7]. Similarly, training with perturbed medical images has been investigated as a means of improving robustness to common perturbations [8]. These studies primarily focus on measuring or improving robustness; they do not directly address whether changes in model behavior can be monitored during degradation without relying on ground-truth labels.

Explainable artificial intelligence provides another potentially informative view of model behavior. Explanation methods such as Grad-CAM can expose the spatial evidence associated with a model's prediction, making it possible to investigate not only whether a prediction changes but also whether the model's visual reasoning changes. Systematic reviews of XAI in medical image analysis have identified methods including LIME, SHAP, Grad-CAM, Integrated Gradients, and layer-wise relevance approaches, while also emphasizing challenges concerning explanation evaluation and robustness [9]. Research has further shown that saliency explanations themselves can be unstable or insufficiently trustworthy. Zhang *et al.* reported weaknesses in the sensitivity and robustness of saliency methods in radiology models [10], while Clark *et al.* demonstrated that post-hoc explanation methods can attribute importance to irrelevant features under controlled benchmark conditions [11]. Studies comparing class-activation-map methods have likewise shown substantial differences in robustness among explanation techniques [12].

More recent work has examined explanation stability directly under input corruption. Repetto *et al.* evaluated multiple post-hoc XAI methods under natural and adversarial corruption and showed that explanations can become unstable even under relatively subtle perturbations [13]. Zhang *et al.* similarly evaluated several explanation approaches over multiple natural image corruptions and severity levels, demonstrating substantial variation in explanation robustness across methods [14]. Marjanović *et al.* investigated the relationship between model instability, explanations, and uncertainty in language models and found that uncertainty and explanation quality do not necessarily vary together [15]. Collectively, these studies establish explanation instability as a measurable property of model behavior. However, explanation instability has generally been treated as a property of explanation quality or robustness rather than being systematically integrated with prediction-level behavioral signals for degradation characterization.

Model degradation and drift monitoring provide a related but broader research direction. Reviews of model degradation emphasize the importance of continuously monitoring deployed AI systems and identifying changes before they result in unacceptable performance [16]. Performance-aware drift detectors such as DDM, EDDM, ADWIN, STEPD, and CUSUM have been studied for detecting changes in model behavior and performance [4]. Recent work has also explored explainability-based monitoring in object detection and poisoning detection, demonstrating that explanation-derived information can provide evidence about changes in model behavior [17], [18]. Nevertheless, most degradation-monitoring approaches either depend on performance feedback or focus on a particular failure mechanism. The combination of multiple prediction-level behavioral signals and explanation instability for progressive image corruption remains comparatively underexplored.

This study addresses this gap by investigating whether heterogeneous behavioral signals can jointly characterize progressive degradation in an image classifier. Rather than assuming that degradation follows a universal sequence, the study explicitly tests the relationship between corruption severity, behavioral changes, explanation instability, and performance failure. A ResNet-50 classifier with 97.53% clean accuracy is evaluated on CIFAR-10-C across 15 corruption types and five severity levels, producing 75 experimental conditions. The proposed Behavioral Degradation Index (BDI) combines five signals: confidence change, Jensen–Shannon divergence between clean and corrupted predictive distributions, prediction flip rate, prediction-margin degradation, and Grad-CAM explanation instability. Each signal is standardized across the complete experimental matrix and combined using equal weights. The implementation therefore treats the BDI as a transparent composite characterization measure rather than a learned predictor. The underlying methodology and signal definitions are described in detail in Section III.

The study investigates four principal questions. First, does Grad-CAM explanation instability systematically change as corruption severity increases? Second, does combining prediction-level and explanation-level signals produce an aggregate BDI that tracks progressive corruption? Third, do different corruption mechanisms follow a common behavioral-to-performance degradation pathway, or do they exhibit distinct pathways? Finally, can adaptive warning thresholds calibrated from mild corruption provide an advance indication of performance failure, including when the evaluated corruption is excluded from threshold calibration?

The main contributions of this study are as follows:

- We formulate a five-signal Behavioral Degradation Index that combines confidence change, Jensen–Shannon divergence, prediction flip rate, prediction-margin degradation, and Grad-CAM explanation instability using transparent equal weighting.
- We evaluate behavioral degradation progressively across 15 CIFAR-10-C corruption types and five severity levels, resulting in 75 corruption–severity conditions rather than relying only on a clean-versus-corrupted comparison.

- We introduce a four-level operational framework distinguishing behavioral shift, decision instability, explanation shift, and performance failure, and use it to characterize degradation pathways across corruption mechanisms.
- We evaluate adaptive Q95 warning thresholds and a leave-one-corruption-out calibration procedure to test whether the proposed signals can provide corruption-independent advance warnings of performance failure.
- We analyze redundancy among the five signals using Pearson correlation and variance inflation factors, followed by multivariate regression and leave-one-signal-out ablation to determine whether explanation instability contributes information distinct from the prediction-based signals.
- We show that the BDI is positively associated with corruption severity, while the primary warning protocol does not provide reliable advance warnings of the defined performance-failure threshold. This negative result provides an explicit empirical baseline for future research on early-warning model-degradation detection.

II. Related Work

A. Failure Detection and Confidence-Based Monitoring

Failure detection attempts to identify predictions that are likely to be incorrect without necessarily requiring the model to be retrained. This problem is particularly important in deployment settings where model predictions may be generated before their true correctness is known. Bernhardt *et al.* evaluated failure detection in medical image classification and showed that conventional confidence measures remain competitive with more sophisticated uncertainty-estimation techniques in several settings [1]. Their results emphasize that uncertainty estimation should be evaluated carefully rather than assuming that greater methodological complexity necessarily produces better failure detection.

Jaeger *et al.* examined evaluation practices for failure detection under distribution shift and argued that failure-detection performance depends strongly on the type of shift and the evaluation protocol [2]. This observation is particularly relevant to corruption-based degradation studies because different corruption mechanisms can produce substantially different changes in model behavior. Bungert *et al.* further studied silent failures in medical image classification using multiple confidence-scoring methods and distribution shifts, demonstrating that no single confidence measure consistently prevents silent failures across all considered shifts [3]. These findings motivate the use of multiple complementary signals rather than relying exclusively on confidence.

The present study extends this perspective by combining confidence change with predictive-distribution divergence, prediction flips, margin degradation, and explanation instability. Unlike conventional failure detection, the objective is not simply to determine whether an individual prediction is incorrect. Instead, the analysis characterizes how the aggregate behavior of a classifier changes as corruption severity increases.

B. Robustness to Common Image Corruptions

Common-corruption benchmarks provide a controlled environment for studying the degradation of image classifiers under systematic input perturbations. Modas *et al.* introduced PRIME, demonstrating that a combination of spectral, color, and spatial augmentation primitives can improve robustness to common corruptions on datasets including CIFAR-10-C and ImageNet-C [5]. The emphasis of such work is primarily on improving the model so that its performance decreases less under corruption.

In medical imaging, Di Salvo *et al.* introduced MedMNIST-C to evaluate classifier robustness under simulated realistic corruptions across multiple medical imaging datasets [6]. Islam *et al.* investigated progressive stress testing across multiple perturbation types and severities and showed that robustness depends on architecture, perturbation type, class, and pretraining [7]. Chuah *et al.* similarly investigated the generation of medical imaging classifiers that are more robust to common perturbations [8].

Although these studies establish the importance of evaluating models progressively under corruption, their primary objective is robustness measurement or robustness improvement. The present work instead asks whether measurable changes in model behavior and explanations can be used to characterize the degradation process itself. CIFAR-10-C is therefore used not simply as a robustness benchmark but as a controlled environment for studying behavioral trajectories across severity levels.

C. Explainability and Reliability of Visual Explanations

Explainable AI has become an important component of trustworthy machine learning because explanations can provide information about the evidence underlying a prediction. In medical image analysis, systematic reviews have examined a wide range of explanation techniques, including LIME, SHAP, Grad-CAM, Integrated Gradients, and relevance-based methods, while identifying the need for stronger and more standardized evaluation of explanation quality [9].

However, the existence of an explanation does not guarantee that the explanation is reliable. Zhang *et al.* investigated the trustworthiness of saliency methods in radiology AI and found that several saliency approaches can exhibit weak sensitivity and robustness under perturbation [10]. Clark *et al.* proposed XAI-TRIS to evaluate false-positive post-hoc feature attribution and showed that explanation methods can assign importance to irrelevant features [11]. Sarkar *et al.* compared multiple class-activation-map variants under noise and proposed a framework based on explanation consistency and responsiveness [12].

These findings suggest that explanation stability should itself be treated as an empirical quantity. In the present study, Grad-CAM instability is explicitly quantified as the mean absolute difference between normalized clean and corrupted explanation maps. Importantly, the clean prediction class is used as the target for both the clean and corrupted images, allowing the experiment to measure how the evidence associated

with the original prediction changes under corruption rather than simply comparing independently selected target classes.

D. Explanation Robustness Under Corruption

Recent research has increasingly examined the behavior of XAI methods under corrupted inputs. Repetto *et al.* evaluated multiple post-hoc explanation methods under natural and adversarial corruption in medical image recognition and reported that explanations can become unstable under perturbations [13]. Zhang *et al.* conducted a broader robustness evaluation across multiple natural image corruptions and severity levels and found substantial differences in the robustness of attribution and perturbation-based explanation methods [14].

Marjanović *et al.* investigated model instability and its relationship with explanations and uncertainty, showing that uncertainty measures and explanation quality do not necessarily exhibit a direct relationship [15]. This distinction is important for the present study because explanation instability is not assumed to be equivalent to prediction uncertainty. Instead, Grad-CAM instability is treated as a separate behavioral dimension and evaluated jointly with prediction-derived signals.

E. Model Degradation and Monitoring

Monitoring degradation after deployment is a broader problem than detecting individual prediction failures. Guan *et al.* reviewed mechanisms for detecting and correcting degradation in medical AI systems and emphasized continuous monitoring as an important component of maintaining trustworthy deployed models [16]. Bayram *et al.* reviewed performance-aware drift detectors and discussed approaches such as DDM, EDDM, ADWIN, STEPD, and CUSUM for detecting changes in model performance [4].

More recent studies have begun incorporating explainability into monitoring. Shin *et al.* investigated early warning of performance degradation in object detection models using explainable AI and demonstrated that explanation-related information can provide useful evidence about changes in model behavior [17]. Asgar *et al.* used explainability-derived features to detect data-poisoning attacks in medical imaging, demonstrating another application in which model explanations provide information beyond conventional output statistics [18].

However, these studies differ from the present work in several respects. First, the present analysis focuses on image classification under controlled progressive corruption rather than object detection or poisoning alone. Second, it combines multiple output-level signals with an explanation-level signal into a single transparent index. Third, it explicitly examines the ordering of behavioral shift, decision instability, explanation shift, and performance failure. Finally, the warning mechanism is evaluated under leave-one-corruption-out calibration to determine whether a threshold calibrated on other corruption mechanisms transfers to an unseen corruption type.

F. Research Gap

The existing literature establishes four important observations. First, confidence and uncertainty can provide useful information about prediction failure, but no single confidence measure is universally reliable under distribution shift [1]–[3]. Second, common corruptions can cause substantial performance degradation, and robustness varies considerably according to corruption type, severity, architecture, and training procedure [5]–[7]. Third, visual explanations can themselves become unstable under corruption and therefore should not automatically be treated as reliable evidence [10], [12]–[14]. Fourth, degradation-monitoring research has increasingly emphasized continuous monitoring, but explanation-derived signals have received comparatively limited attention in classification settings [4], [16], [17].

Despite these developments, there remains a need for controlled experimental analysis of how multiple behavioral dimensions evolve jointly under progressive image corruption. In particular, it is not yet clear whether output-level signals such as confidence change, predictive-distribution divergence, prediction flips, and margin degradation provide information that is meaningfully complemented by explanation instability. It is also unclear whether these signals follow a common degradation pathway before performance failure or whether different corruption mechanisms produce fundamentally different sequences of behavioral change.

This study therefore does not assume that behavioral shift necessarily precedes performance failure. Instead, it formulates this as an empirical question and evaluates it using a fixed ResNet-50 model, 15 CIFAR-10-C corruption types, five severity levels, adaptive Q95 thresholds, and leave-one-corruption-out validation. The resulting analysis distinguishes between three related but different outcomes: *association with degradation*, *characterization of degradation pathways*, and *advance warning of performance failure*. This distinction is central to interpreting the experimental findings. The results show that the BDI is positively associated with corruption severity, while the primary warning protocol does not reliably provide advance warning, establishing a useful baseline rather than an unsupported claim of successful early-warning detection.

III. Methodology

This study investigates the relationship between corruption severity, model behavior, decision instability, explanation instability, and classification performance in an image classification model. The experimental framework combines four prediction-based degradation signals with a Grad-CAM-based explanation-instability signal to form a Behavioral Degradation Index (BDI). The resulting index is evaluated across multiple corruption types and severity levels, followed by adaptive-threshold analysis, multilevel degradation-pathway analysis, early-warning analysis, leave-one-corruption-out (LOCO) validation, multicollinearity analysis, regression, and leave-one-signal-out ablation.

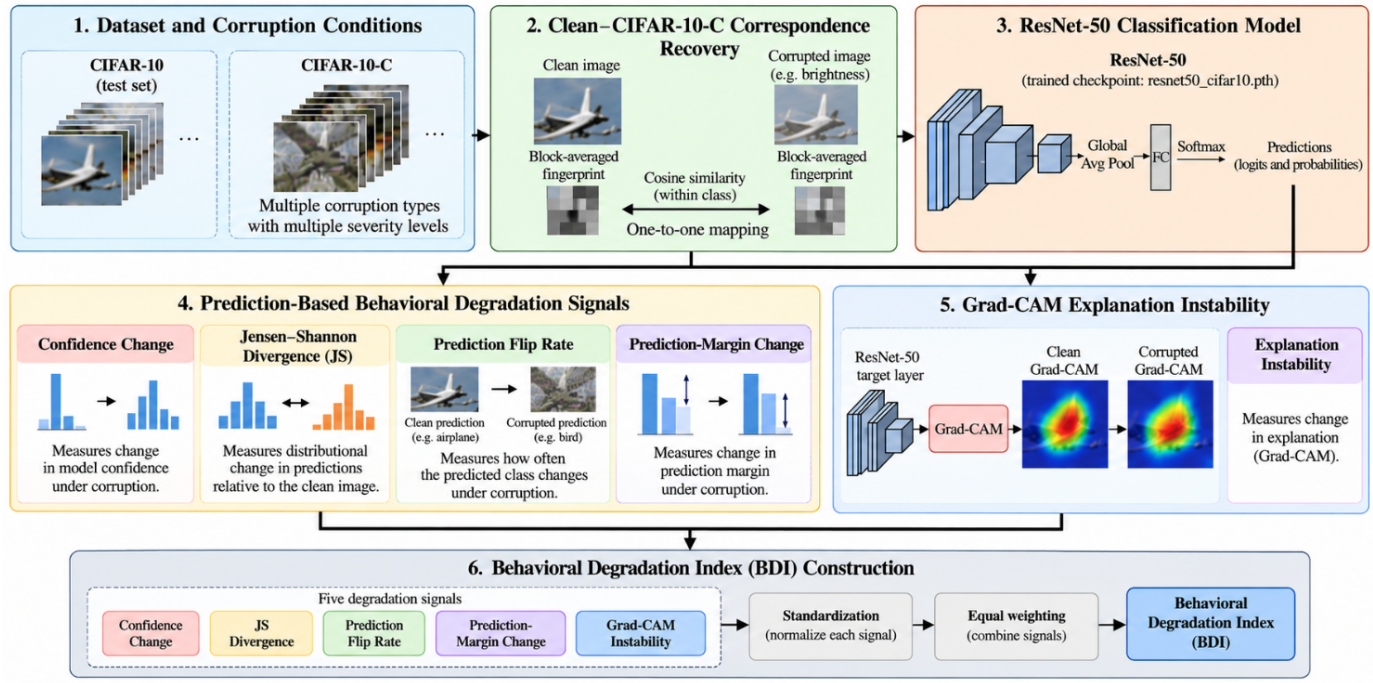


Fig. 1. Multilevel degradation framework used in the study.

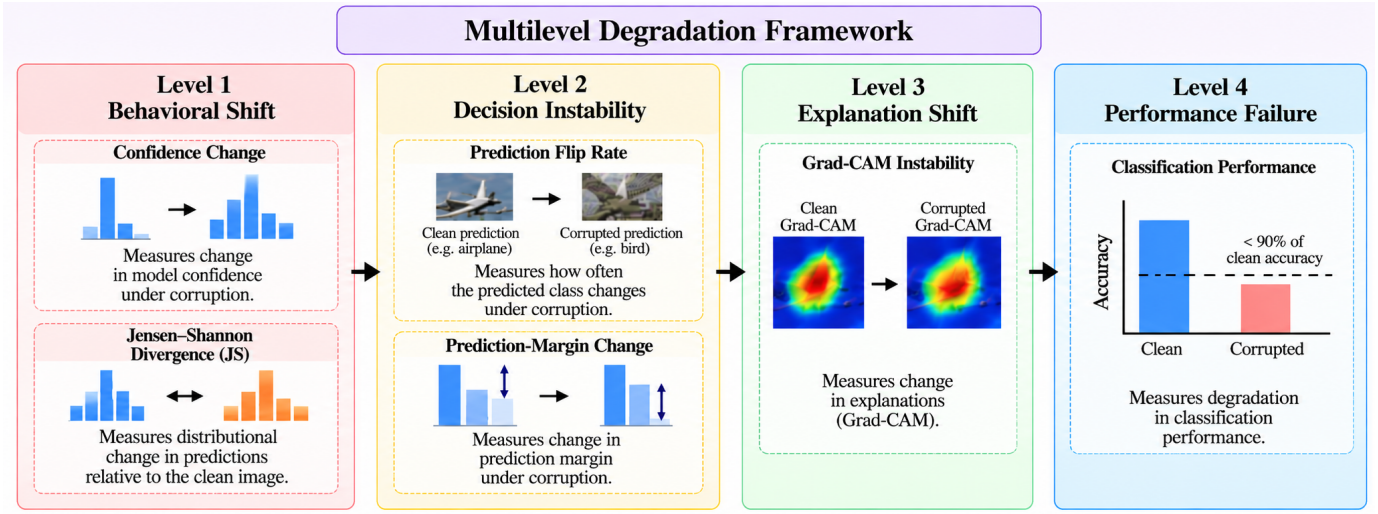


Fig. 2. Overall experimental methodology of the study.

A. Experimental Dataset and Corruption Conditions

The experiments use the CIFAR-10 test set together with the CIFAR-10-C corruption benchmark. The clean CIFAR-10 test set contains 10,000 images distributed across the ten CIFAR-10 classes. The ten classes are airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

Fifteen CIFAR-10-C corruption types are considered: brightness, contrast, defocus blur, elastic transformation, fog, frost, Gaussian blur, Gaussian noise, glass blur, impulse noise, JPEG compression, motion blur, pixelation, saturation, and shot noise. Each corruption contains five severity levels. Thus, the complete experimental matrix contains

$$15 \times 5 = 75 \tag{1}$$

corruption–severity conditions.

For every condition, the analysis operates on 10,000 corresponding clean/corrupted image pairs. The behavioral signals, explanation instability, and classification accuracy are subsequently aggregated at the corruption–severity level.

B. Recovery of Clean–CIFAR-10-C Correspondence

The Grad-CAM analysis requires a correspondence between each CIFAR-10-C image and its clean CIFAR-10 counterpart.

The notebook recovers this correspondence using the severity-1 brightness corruption and image fingerprints.

Each clean and corrupted image is represented by an $8 \times 8 \times 3$ fingerprint obtained from the original $32 \times 32 \times 3$ image using 4×4 block averaging. For each image, the fingerprint is standardized using

$$F' = \frac{F - \mu_F}{\sigma_F + 10^{-6}}, \tag{2}$$

where F is the block-averaged fingerprint and μ_F and σ_F are calculated over the spatial and channel dimensions of that image.

The matching procedure is performed independently within each CIFAR-10 class. The flattened fingerprints are L_2-normalized and compared using cosine similarity:

$$S(a, b) = \frac{a^\top b}{\|a\|_2 \|b\|_2}. \tag{3}$$

For each CIFAR-10-C severity-1 brightness sample, the clean image with the highest cosine similarity within the corresponding class is selected. The resulting mapping contains 10,000 entries and is checked for one-to-one correspondence. This mapping is subsequently used to retrieve the clean counterpart for every corruption and severity condition.

C. ResNet-50 Classification Model

A ResNet-50 architecture is used as the classification model. The network is instantiated without ImageNet pretrained weights and its final fully connected layer is configured for ten output classes:

$$f_\theta(x) \in \mathbb{R}^{10}. \tag{4}$$

The trained checkpoint `resnet50_cifar10.pth` is loaded into the model. The checkpoint loader supports both direct state dictionaries and checkpoints containing a `model_state_dict` or `state_dict` field. A possible `module.` prefix is removed before loading the parameters.

The checkpoint-reported clean model accuracy used throughout the primary analysis is

$$A_{\text{clean}} = 0.975300, \tag{5}$$

corresponding to 97.53% clean accuracy. This value is treated as a single fixed analysis constant throughout the downstream calculations, including the performance-failure threshold.

D. Grad-CAM Explanation Instability

1) Input Preprocessing: The corrected Grad-CAM experiment uses the preprocessing pipeline specified in the experimental code. Each 32×32 RGB image is converted to a floating-point tensor, scaled from $[0, 255]$ to $[0, 1]$, resized to 224×224 using bilinear interpolation, and normalized using the ImageNet mean and standard deviation:

$$\boldsymbol{\mu} = (0.485, 0.456, 0.406), \tag{6}$$

$$\boldsymbol{\sigma} = (0.229, 0.224, 0.225). \tag{7}$$

The corrected Grad-CAM experiment uses a batch size of 64 and evaluates all 15 corruption types and all five severity levels. The preprocessing pipeline is applied identically to clean and corrupted images; therefore, Grad-CAM instability is interpreted as a relative explanation-change measure under the specified preprocessing rather than as a direct measure of the classifier's native training-time explanation behavior.

2) Target Layer: The Grad-CAM target is the final block of the third residual stage:

$$L = \texttt{model.layer3[-1]}. \tag{8}$$

This layer is used consistently for the complete Grad-CAM experiment.

3) Grad-CAM Computation: Let A^k denote the activation map of channel k at the selected target layer and let y^c denote the logit corresponding to target class c. The Grad-CAM channel weight is calculated by global average pooling of the spatial gradients:

$$\alpha_k^c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \frac{\partial y^c}{\partial A_{ij}^k}. \tag{9}$$

The class activation map is then computed as

$$CAM^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right). \tag{10}$$

The resulting CAM is resized to a common 224×224 resolution using bilinear interpolation. Each CAM is independently normalized to the range $[0, 1]$:

$$\widehat{CAM} = \frac{CAM - CAM_{\min}}{CAM_{\max} - CAM_{\min} + 10^{-8}}. \tag{11}$$

4) Target-Class Consistency: For each paired clean/corrupted sample, the predicted class obtained from the clean image is used as the Grad-CAM target for both the clean and corrupted images. Thus, the explanation comparison measures how the spatial explanation associated with the original clean prediction changes under corruption.

This procedure avoids selecting independent target classes for the clean and corrupted images when computing explanation instability.

5) Explanation Instability: For each paired image, explanation instability is calculated as the mean absolute difference between the normalized clean and corrupted Grad-CAM maps:

$$G_i = \frac{1}{HW} \sum_{u=1}^{H} \sum_{v=1}^{W} \left| \widehat{CAM}_{i,\text{clean}}(u, v) - \widehat{CAM}_{i,\text{corr}}(u, v) \right|. \tag{12}$$

For each corruption–severity condition, the mean explanation instability is calculated over the 10,000 paired samples:

$$G_{c,s} = \frac{1}{N}\sum_{i=1}^{N} G_i, \qquad N = 10000. \tag{13}$$

The corrected Grad-CAM experiment additionally records the median, standard deviation, 90th percentile, and 95th percentile of the per-image instability values. The mean instability is the quantity used as the Grad-CAM component of the BDI.

E. Prediction-Based Degradation Signals

Four prediction-based signals are obtained from the previously generated prediction and margin analyses.

1) Confidence Change: The confidence signal is obtained from the mean confidence change between the clean and corrupted inputs. The signal is denoted by C. The implementation checks the direction of the resulting confidence-change statistic. If its overall mean is negative, the signal is multiplied by -1 so that the resulting BDI input follows the intended degradation direction, where larger values represent greater degradation.

2) Jensen–Shannon Divergence: For each paired sample, the clean and corrupted predictive distributions are compared using Jensen–Shannon divergence. For distributions P and Q,

$$M = \frac{1}{2}(P + Q), \tag{14}$$

and

$$JS(P, Q) = \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M). \tag{15}$$

The condition-level mean is denoted by J.

3) Prediction Flip Rate: Prediction instability is measured using the fraction of paired samples for which the predicted class changes between the clean and corrupted inputs. For N samples,

$$F = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left[\hat{y}_{i,\text{clean}} \neq \hat{y}_{i,\text{corr}}\right]. \tag{16}$$

The resulting condition-level quantity is denoted by F.

4) Prediction-Margin Change: The prediction-margin signal is obtained from the previously generated margin analysis. For each condition, the quantity `delta_margin_mean` is used as the margin-degradation signal and is denoted by M_Δ. Larger values are interpreted as greater margin degradation according to the direction used by the BDI analysis.

F. Construction of the Behavioral Degradation Index

The five signals used to construct the BDI are

$$\mathbf{s} = [C, J, F, M_\Delta, G]. \tag{17}$$

Before aggregation, each signal is converted to a standardized z-score using its mean and sample standard deviation calculated across the 75 corruption–severity conditions:

$$Z_s = \frac{s - \mu_s}{\sigma_s}, \tag{18}$$

where μ_s and σ_s are calculated from the corresponding signal over the complete 75-condition BDI dataset.

The five standardized signals are combined using equal weights:

$$\boxed{BDI = 0.20Z_C + 0.20Z_J + 0.20Z_F + 0.20Z_{M_\Delta} + 0.20Z_G} \tag{19}$$

or equivalently,

$$BDI = \frac{1}{5}\sum_{s\in\mathcal{S}} Z_s. \tag{20}$$

The equal weighting is used as a transparent fixed aggregation rule; the weights are not learned from the data. Because the primary BDI standardization uses the complete 75-condition matrix, the primary BDI is treated as a descriptive characterization measure rather than a deployment-ready predictive score. The held-out LOCO analysis instead uses development-only statistics for calibration.

For visualization, the BDI is additionally transformed to the $[0, 1]$ range using

$$BDI_{0-1} = \frac{BDI - BDI_{\min}}{BDI_{\max} - BDI_{\min}}. \tag{21}$$

The standardized BDI is retained for the primary BDI analyses.

G. BDI–Severity Validation

The relationship between BDI and corruption severity is evaluated using Pearson correlation, Spearman rank correlation, and Kendall's rank correlation. These statistics are treated as descriptive association measures because the 75 condition-level observations are nested within 15 corruption types, with five severity levels per corruption. No independence-based significance test is used for the primary correlation interpretation.

For the global descriptive analysis, all 75 corruption–severity observations are used. If s_i denotes the severity and BDI_i denotes the corresponding BDI value, Pearson correlation is used to characterize the linear relationship, while Spearman and Kendall statistics are used to characterize monotonic association. Because repeated severity observations belong to the same corruption type, these coefficients are not interpreted as independent-observation hypothesis tests.

The mean BDI is additionally calculated separately for each of the five severity levels. For each corruption type, the five BDI values are independently ordered by severity and evaluated using Spearman correlation.

For each corruption, the fraction of adjacent severity transitions that increase BDI is also calculated:

$$M_c = \frac{\sum_{s=1}^{4} \mathbb{I}(BDI_{c,s+1} > BDI_{c,s})}{4}. \tag{22}$$

The mean of M_c across the 15 corruption types is used as the corruption-wise monotonicity measure.

The BDI is also compared with each individual degradation signal using Spearman correlation, Kendall correlation, and the monotonicity of their severity-wise means.

H. Adaptive Warning Thresholds

The adaptive warning thresholds are calibrated from the condition-level severity-1 observations. Severity 1 is used as the mild-degradation reference condition. Because there are 15 corruption types, the primary reference distribution contains $n = 15$ condition-level observations for each monitored quantity. The resulting percentile thresholds are therefore empirical, small-sample calibration values and are interpreted as exploratory warning thresholds rather than precise estimates of a population 95th percentile.

For each of the six monitored quantities

$$\{C, J, F, M_\Delta, G, BDI\},$$

three percentile thresholds are calculated from the severity-1 reference distribution:

$$T_{90} = Q_{0.90}, \tag{23}$$

$$T_{95} = Q_{0.95}, \tag{24}$$

and

$$T_{99} = Q_{0.99}. \tag{25}$$

The Q95 threshold is used as the primary threshold in the main multilevel and early-warning analyses. Q90 and Q99 are used for threshold-sensitivity analysis. The small reference-set size is kept explicit so that the warning results are not overstated.

For the main threshold-crossing diagnostics, a warning is recorded when the monitored quantity reaches or exceeds its Q95 threshold:

$$s \geq T_{95}. \tag{26}$$

The same $\geq$ convention is used in the multilevel threshold rules defined below.

I. Operational Definition of Performance Failure

Performance failure is defined relative to the clean model accuracy. The primary failure threshold is 90% of clean accuracy:

$$T_{\text{failure}} = 0.90 A_{\text{clean}}. \tag{27}$$

Using

$$A_{\text{clean}} = 0.975300, \tag{28}$$

the primary threshold is

$$T_{\text{failure}} = 0.90(0.975300) = 0.877770. \tag{29}$$

A condition is therefore classified as a performance failure when

$$Accuracy < T_{\text{failure}}. \tag{30}$$

For each corruption type, the first severity satisfying this condition is recorded as the performance-failure severity.

Two additional failure definitions are evaluated for sensitivity analysis:

$$T_{95} = 0.95 A_{\text{clean}}, \tag{31}$$

and

$$T_{85} = 0.85 A_{\text{clean}}. \tag{32}$$

The 90%-of-clean threshold is the primary operational definition.

J. Multilevel Degradation Framework

The study defines four operational degradation levels based on the Q95 thresholds.

1) Level 1: Behavioral Shift: Level 1 is reached when either the confidence signal or JS divergence crosses its Q95 threshold:

$$L_1 = (C \geq T_C) \vee (J \geq T_J). \tag{33}$$

2) Level 2: Decision Instability: Level 2 is reached when either prediction flip rate or margin degradation crosses its Q95 threshold:

$$L_2 = (F \geq T_F) \vee (M_\Delta \geq T_M). \tag{34}$$

3) Level 3: Explanation Shift: Level 3 is reached when Grad-CAM instability crosses its Q95 threshold:

$$L_3 = (G \geq T_G). \tag{35}$$

4) Level 4: Performance Failure: Level 4 is reached when accuracy falls below the primary performance threshold:

$$L_4 = (Accuracy < T_{\text{failure}}). \tag{36}$$

The conceptual degradation hierarchy investigated in the study is

$$L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4. \tag{37}$$

For each corruption type, the first severity at which each level is reached is recorded.

K. Degradation-Pathway Classification

For every corruption, the first severity associated with each reachable level is extracted. The observed order of the levels is obtained by sorting their first-reached severities, with the level number used as the secondary ordering criterion when two levels are reached at the same severity.

Each corruption is subsequently assigned to one of the following categories:

1) *Complete* $L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4$: all four levels are reached and satisfy
$$s_1 \leq s_2 \leq s_3 \leq s_4.$$
2) *Non-decreasing with skip*: multiple levels are observed and their first-reached severities are non-decreasing, but one or more intermediate levels are absent.
3) *Reversal/non-monotonic*: multiple levels are observed but their first-reached severities are not non-decreasing.
4) *Single level*: only one of the four levels is reached.
5) *No levels reached*: none of the four levels is reached.

The first detected level for each corruption is also recorded by selecting the level with the smallest first-reached severity, with level number used as the tie-breaking criterion.

L. Early-Warning Analysis

The primary early-warning analysis evaluates whether a degradation signal crosses its warning threshold before the model reaches the defined performance-failure threshold.

For a signal s, the first warning severity for corruption c is

$$s_{W,c} = \min\{s : s_{c,s} > T_s\}. \tag{38}$$

The first failure severity is

$$s_{F,c} = \min\{s : Accuracy_{c,s} < T_{\text{failure}}\}. \tag{39}$$

The severity-based lead time is

$$LeadTime_c = s_{F,c} - s_{W,c}. \tag{40}$$

When a corruption reaches the failure threshold, the outcome is classified as follows:

- *Early warning* if $s_{W,c} < s_{F,c}$;
- *Warning at failure* if $s_{W,c} = s_{F,c}$;
- *Late warning* if $s_{W,c} > s_{F,c}$;
- *Missed warning* if no warning is observed before or at failure.

When no performance failure is observed for a corruption, the trajectory is treated separately as having no observed failure.

The analysis is repeated for Confidence, JS, Flip, Margin, GradCAM, and BDI. Q95 is the primary warning threshold, while Q90 and Q99 are used in the threshold-sensitivity analysis.

M. Leave-One-Corruption-Out Validation

A leave-one-corruption-out (LOCO) procedure is used to evaluate whether the BDI warning mechanism generalizes to a corruption type that is excluded from calibration.

For each of the 15 corruption types, one corruption is designated as the held-out test corruption and the remaining 14 corruption types are used as the development set.

For each LOCO fold, the severity-1 observations of the 14 development corruptions constitute the reference set. Thus, each LOCO calibration uses only $n = 14$ condition-level reference observations for each BDI component. For every BDI component s, its development reference mean and sample standard deviation are calculated as

$$\mu_{s,\text{dev}} = \text{mean}\left(s_{\text{dev},\, severity=1}\right), \tag{41}$$

$$\sigma_{s,\text{dev}} = \text{std}\left(s_{\text{dev},\, severity=1}\right). \tag{42}$$

The held-out corruption is standardized using only these development statistics:

$$Z_{s,\text{test}} = \frac{s_{\text{test}} - \mu_{s,\text{dev}}}{\sigma_{s,\text{dev}}}. \tag{43}$$

The LOCO BDI is then calculated using the fixed equal weights:

$$BDI_{\text{LOCO}} = \frac{1}{5}\sum_{s\in\mathcal{S}} Z_{s,\text{test}}. \tag{44}$$

A separate development-reference BDI distribution is calculated using the same development means and standard deviations. It contains 14 severity-1 development observations. The LOCO warning threshold is its 95th percentile:

$$T_{\text{BDI,LOCO}} = Q_{0.95}\left(BDI_{\text{dev,reference}}\right). \tag{45}$$

The held-out corruption is considered to generate a BDI warning when

$$BDI_{\text{LOCO}} \geq T_{\text{BDI,LOCO}}. \tag{46}$$

The same primary performance-failure definition, $Accuracy < 0.90A_{\text{clean}}$, is then used to determine whether the warning occurs before, at, or after performance failure.

N. Multicollinearity Analysis

The relationships among the five BDI component signals are examined using a Pearson correlation matrix.

Variance Inflation Factor (VIF) is used to quantify multicollinearity. For predictor j,

$$VIF_j = \frac{1}{1 - R_j^2}, \tag{47}$$

where R_j^2 is obtained by regressing predictor j against the remaining degradation signals.

This analysis is used to characterize redundancy among the component signals and to determine whether individual coefficients in the subsequent multivariate regression should be interpreted cautiously.

O. Five-Signal Regression Analysis

A multiple ordinary least-squares regression is fitted as a descriptive analysis of the relationship between the five degradation signals and accuracy degradation. The regression uses the 75 corruption–severity condition-level observations, which are clustered within 15 corruption types. Consequently, the fitted coefficients and R^2 values are used to describe in-sample association and signal redundancy rather than to make independent-observation inferential or causal claims.

Accuracy degradation is defined as

$$D_A = A_{\text{clean}} - A_{\text{corr}}, \tag{48}$$

where A_{corr} is the accuracy associated with a particular corruption–severity condition.

The regression model is

$$D_A = \beta_0 + \beta_1 C + \beta_2 J + \beta_3 F + \beta_4 M_\Delta + \beta_5 G + \epsilon. \tag{49}$$

The fitted model is evaluated using R^2 and adjusted R^2 as descriptive goodness-of-fit measures. Standard errors, t statistics, and p values are not used for inferential claims because the 75 observations are clustered by corruption type and the predictors are strongly redundant.

Because the five degradation signals are evaluated jointly and may be strongly correlated, the regression coefficients are interpreted together with the VIF analysis as descriptive association coefficients rather than as independent causal effects.

P. Leave-One-Signal-Out Ablation

Two complementary leave-one-signal-out analyses are performed.

First, the contribution of each component to the BDI's association with corruption severity is evaluated by removing one standardized signal at a time and recomputing the BDI as the arithmetic mean of the remaining four standardized signals.

For a removed signal r,

$$BDI_{-r} = \frac{1}{4} \sum_{s \in \mathcal{S} \setminus \{r\}} Z_s. \tag{50}$$

The Spearman correlation, Kendall correlation, and mean severity-wise monotonicity of each ablated index are compared with those of the complete five-signal BDI. The change in Spearman correlation is calculated as

$$\Delta\rho_r = \rho_{-r} - \rho_{\text{BDI}}. \tag{51}$$

Second, a leave-one-signal-out regression analysis is performed. For each signal, that signal is removed from the five-signal regression, the remaining four signals are fitted using ordinary least squares, and the resulting R^2, adjusted R^2, and change in R^2 relative to the complete model are recorded.

Q. Data Validation and Experimental Consistency

Before the multilevel analysis, the BDI, JS, margin, Grad-CAM, and accuracy data are merged using the corruption and severity identifiers. The implementation requires one-to-one correspondence for these keys and verifies that the merged dataset contains exactly 75 corruption–severity conditions. The condition-level accuracy table used by the downstream LOCO and regression analyses is explicitly constructed from the same condition-level results before those analyses, so no undeclared external accuracy variable is required.

Duplicate corruption–severity combinations are checked before merging, and missing signal values are checked during BDI construction. The Grad-CAM experiment additionally verifies the 10,000-entry clean–CIFAR-10-C mapping, the 224×224 input resolution, the Grad-CAM output resolution, and the successful execution of the Grad-CAM sanity test.

The resulting framework therefore uses a fixed ResNet-50 checkpoint, fixed preprocessing for the corrected Grad-CAM analysis, fixed equal BDI weights, explicitly defined percentile thresholds, and LOCO-specific calibration for held-out-corruption evaluation.

IV. Results

The experiments were conducted over 15 CIFAR-10-C corruption types and five severity levels, yielding 75 corruption–severity conditions. The clean ResNet-50 achieved 97.53% accuracy, corresponding to a primary performance-failure threshold of 87.777%, defined as 90% of clean accuracy. The results are organized around four questions: (i) whether explanation instability increases under corruption, (ii) whether the proposed Behavioral Degradation Index (BDI) tracks corruption severity, (iii) whether the proposed multilevel degradation hierarchy is observed consistently across corruption types, and (iv) whether the resulting signals provide corruption-independent warning of performance failure.

A. Explanation Instability Under Corruption

The corrected Grad-CAM experiment was successfully evaluated across all 15 corruption types and five severity levels. The resulting explanation-instability values demonstrate substantial variation across corruption families and generally increase with corruption severity. Figure 3 summarizes the mean Grad-CAM instability for all 75 corruption–severity conditions.

Several corruption types produced comparatively large explanation instability even at low severity. For example, Gaussian noise, glass blur, impulse noise, JPEG compression, and shot noise exhibited higher instability at severity 1 than brightness, contrast, or blur-based corruptions. Gaussian noise showed a particularly strong progression, with mean Grad-CAM instability increasing from approximately 0.155 at severity 1 to 0.200 at severity 5. Impulse noise similarly increased from approximately 0.144 to 0.197. In contrast, brightness produced a more gradual increase, from approximately 0.048 at severity 1 to 0.122 at severity 5.

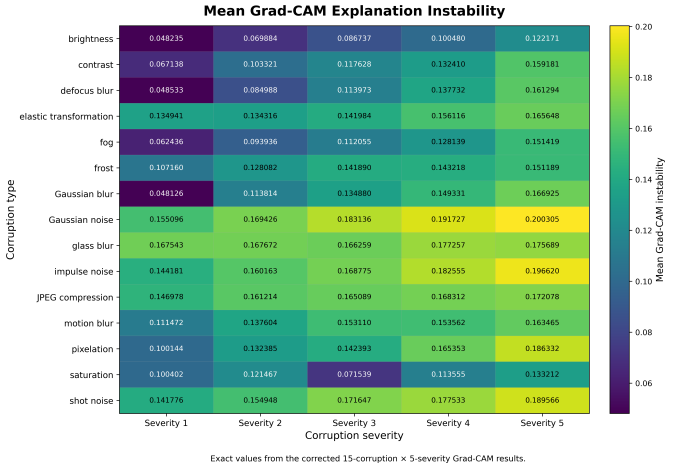


Fig. 3. Mean Grad-CAM explanation instability across the 15 CIFAR-10-C corruption types and five severity levels. Each cell represents the mean absolute difference between normalized clean and corrupted Grad-CAM maps over the 10,000 paired images for the corresponding corruption–severity condition.

TABLE I
SEVERITY-WISE BEHAVIORAL DEGRADATION INDEX PROFILE.

Severity	Mean	Median	Std. Dev.	Minimum	Maximum
1	-0.6517	-0.8267	0.5701	-1.2313	0.6945
2	-0.3244	-0.6010	0.6610	-1.0993	1.0522
3	-0.0622	-0.3342	0.8711	-1.0912	1.8645
4	0.2879	-0.0827	1.0133	-0.8754	2.2122
5	0.7504	0.3868	1.0329	-0.6799	2.5120

These results indicate that corruption can alter the spatial evidence used by the classifier even when the magnitude of the change is strongly dependent on corruption type. Thus, explanation instability captures a behavioral dimension that is not reducible to a single universal severity response.

B. Severity-Dependent Behavioral Degradation

The proposed BDI was evaluated against corruption severity using Pearson, Spearman, and Kendall correlation. A positive association was observed for all three measures. Across the complete set of 75 condition-level observations, Pearson correlation was $r = 0.5063$, Spearman correlation was $\rho = 0.5239$, and Kendall correlation was $\tau = 0.3962$. These coefficients are reported descriptively because the five severity observations for each corruption type are not independent.

The severity-wise mean BDI increased progressively from -0.6517 at severity 1 to 0.7504 at severity 5 (Table I). The corresponding mean values were -0.6517, -0.3244, -0.0622, 0.2879, and 0.7504 for severities 1 through 5, respectively. Thus, the aggregate BDI exhibits a clear upward trend as corruption becomes more severe.

Figure 4 further shows the individual corruption trajectories. Although the magnitude of BDI differs substantially among corruption types, the overall direction is toward greater degradation with increasing severity. The corresponding BDI heatmap in Figure 5 illustrates the same heterogeneity in a condition-level representation. High positive BDI values are

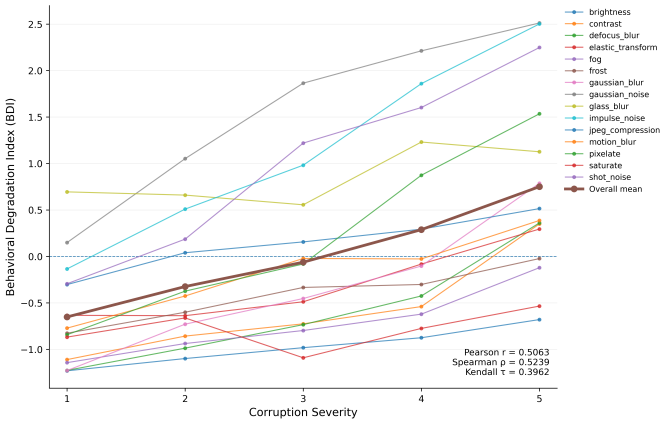


Fig. 4. Behavioral Degradation Index as a function of corruption severity. Thin trajectories represent individual corruption types, while the emphasized trajectory represents the overall severity-wise mean. Pearson, Spearman, and Kendall associations are shown in the lower-right corner.

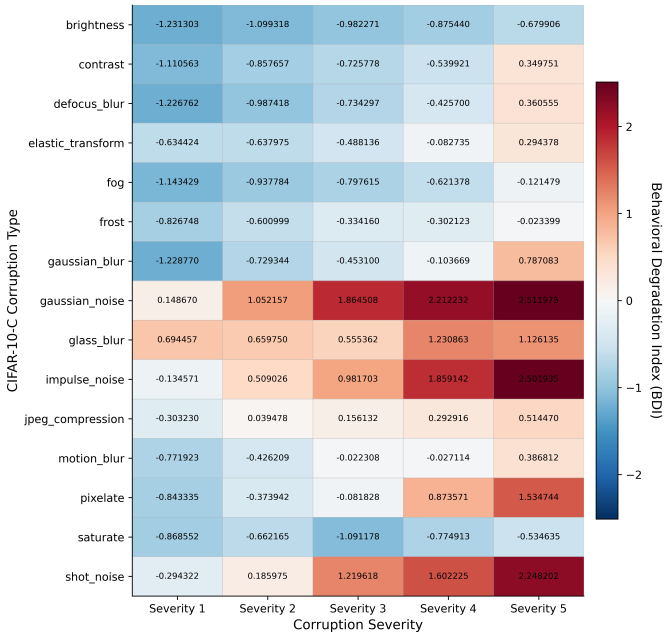


Fig. 5. Behavioral Degradation Index across the 15 corruption types and five severity levels. Values are the standardized five-signal BDI for each corruption–severity condition.

concentrated particularly in Gaussian noise, impulse noise, shot noise, pixelation, and glass blur at higher severities, whereas several corruptions remain below the overall mean even at severity 5.

C. Component-Level Degradation Patterns

The five BDI components also showed positive associations with corruption severity. Grad-CAM instability produced the strongest individual Spearman association with severity ($\rho = 0.5387$), followed by prediction flip rate ($\rho = 0.5287$), JS divergence ($\rho = 0.5278$), BDI ($\rho = 0.5239$), margin degradation ($\rho = 0.5078$), and confidence change ($\rho = 0.5074$).

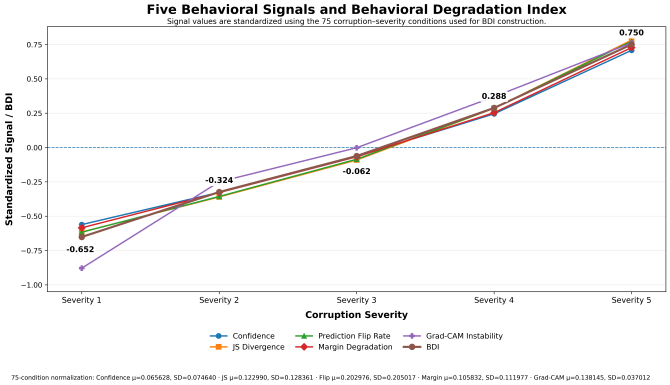


Fig. 6. Severity-wise standardized trajectories of the five degradation signals and the aggregate BDI. The visualization is used to compare the relative severity profiles of the component signals; the primary BDI statistics are calculated from the 75-condition standardization defined in the methodology.

TABLE II
DESCRIPTIVE ASSOCIATION OF INDIVIDUAL DEGRADATION SIGNALS WITH CORRUPTION SEVERITY. THE CORRELATION COEFFICIENTS ARE NOT INTERPRETED USING INDEPENDENCE-BASED SIGNIFICANCE TESTS BECAUSE THE OBSERVATIONS ARE CLUSTERED BY CORRUPTION TYPE.

Signal	Spearman ρ	Kendall τ	Severity-wise Monotonicity
Grad-CAM	0.5387	0.4114	1.00
Flip	0.5287	0.4027	1.00
JS	0.5278	0.4034	1.00
BDI	0.5239	0.3962	1.00
Margin	0.5078	0.3842	1.00
Confidence	0.5074	0.3842	1.00

All five individual signals exhibited positive corruption-wise severity associations, with all 15 corruption types having positive Spearman correlations. The mean adjacent-severity monotonicity of the BDI was 0.9000, indicating that 90% of adjacent severity transitions increased the BDI when averaged across corruption types.

Figure 6 compares the severity-wise standardized behavior of the five component signals and the aggregate index. The curves demonstrate that the component signals broadly move in the same direction as severity, while Grad-CAM exhibits a somewhat different profile from the prediction-based signals. This distinction becomes important in the subsequent multi-collinearity and ablation analyses.

D. Multilevel Degradation Pathways

The four-level framework was next examined using the adaptive Q95 thresholds for behavioral, decision, and explanation signals together with the 90%-of-clean-accuracy threshold for performance failure.

The reachability of the four levels differed across corruption types. Level 1 was reached by 10 of the 15 corruptions (66.7%), Level 2 by 10 (66.7%), Level 3 by 11 (73.3

The complete first-reached severity results are presented in Table III. Only contrast, defocus blur, and glass blur followed the complete $L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4$ sequence. In contrast, eight corruption types exhibited non-monotonic ordering of the first-reached levels. Gaussian blur, Gaussian noise, impulse noise, motion blur, pixelation, and shot noise, for example, reached performance failure at severity 1–3 before the behavioral, decision, and explanation thresholds were subsequently crossed.

TABLE III
FIRST SEVERITY AT WHICH EACH MULTILEVEL DEGRADATION CRITERION WAS REACHED. "–" DENOTES THAT THE CORRESPONDING LEVEL WAS NOT REACHED.

Corruption	L1	L2	L3	L4	Observed pathway
Brightness	–	–	–	–	None
Contrast	5	5	5	5	$L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4$
Defocus blur	5	5	5	5	$L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4$
Elastic transform	–	–	5	4	$L_4 \rightarrow L_3$
Fog	–	–	–	5	L_4
Frost	–	–	–	4	L_4
Gaussian blur	5	5	5	4	$L_4 \rightarrow L_1 \rightarrow L_2 \rightarrow L_3$
Gaussian noise	2	2	2	1	$L_4 \rightarrow L_1 \rightarrow L_2 \rightarrow L_3$
Glass blur	1	1	1	1	$L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4$
Impulse noise	2	2	2	1	$L_4 \rightarrow L_1 \rightarrow L_2 \rightarrow L_3$
JPEG compression	4	4	2	2	$L_3 \rightarrow L_4 \rightarrow L_1 \rightarrow L_2$
Motion blur	5	5	5	3	$L_4 \rightarrow L_1 \rightarrow L_2 \rightarrow L_3$
Pixelation	4	4	4	3	$L_4 \rightarrow L_1 \rightarrow L_2 \rightarrow L_3$
Saturation	–	–	–	–	None
Shot noise	3	3	3	2	$L_4 \rightarrow L_1 \rightarrow L_2 \rightarrow L_3$

JPEG compression exhibited another reversal pattern, with explanation shift and performance failure occurring before the behavioral and decision thresholds. Elastic transformation reached L_3 and L_4 but did not reach L_1 or L_2, whereas fog and frost reached only performance failure. Brightness and saturation did not reach any of the four defined levels.

Overall, 3/15 corruptions (20.0%) followed the complete conceptual sequence, 8/15 (53.3%) exhibited reversal/non-monotonic progression, 2/15 (13.3%) reached only a single level, and 2/15 (13.3%) reached no level. No corruption was classified as a non-decreasing pathway with an intermediate level skipped.

E. Adaptive Warning Performance

The adaptive warning thresholds were calibrated from the 15 condition-level severity-1 observations. The Q95 thresholds used in the primary analysis were 0.0861 for confidence, 0.1485 for JS divergence, 0.2432 for flip rate, 0.1433 for margin degradation, 0.1588 for Grad-CAM instability, and 0.3124 for the BDI. Because each threshold is estimated from only 15 condition-level reference values, these thresholds should be interpreted as exploratory calibration values rather than stable population quantiles.

The primary performance-failure threshold was 0.87777, corresponding to 90% of the 0.97530 clean accuracy. Thirteen of the 15 corruption types reached this threshold at some severity. Brightness and saturation did not exhibit an observed failure under this definition.

Despite the positive severity association of the BDI, none of the six monitored quantities produced an early warning at the primary Q95 threshold. The BDI generated three warnings at the same severity as failure, seven late warnings, and three missed warnings among the 13 corruption types with observed failure. Confidence, JS, flip rate, and margin showed the same

TABLE IV
PRIMARY ADAPTIVE EARLY-WARNING RESULTS USING Q95 WARNING THRESHOLDS AND THE 90%-OF-CLEAN-ACCURACY FAILURE CRITERION.

Signal	Q95 threshold	Observed failures	Early	At failure	Late / Missed
Confidence	0.0861	13	0	3	6 / 4
JS	0.1485	13	0	3	7 / 3
Flip	0.2432	13	0	3	7 / 3
Margin	0.1433	13	0	3	6 / 4
Grad-CAM	0.1588	13	0	4	7 / 2
BDI	0.3124	13	0	3	7 / 3

absence of early warnings, while Grad-CAM produced four warnings at failure, seven late warnings, and two missed warnings.

Thus, the severity association of BDI should not be interpreted as evidence of advance-warning capability. Under the present adaptive Q95 calibration, the BDI detects degradation at or after the defined performance-failure point rather than reliably preceding it.

The primary warning thresholds are based on only 15 severity-1 condition-level reference observations. Accordingly, the warning counts should be interpreted as empirical results for this controlled benchmark rather than as precise estimates of a deployment-level false-alarm or detection rate.

F. Leave-One-Corruption-Out Generalization

To determine whether the warning behavior depended on calibration using the same corruption being evaluated, the BDI warning mechanism was recalibrated using a leave-one-corruption-out procedure. In each fold, the severity-1 observations of the other 14 corruptions were used to estimate the signal means and standard deviations, and the held-out corruption was then evaluated using these development-only statistics. Thus, each LOCO threshold is calibrated from only 14 condition-level severity-1 observations, making the LOCO analysis a held-out generalization check with a small empirical calibration set rather than a high-precision estimate of a population warning threshold.

Figure 7 shows the resulting LOCO BDI trajectories and development-set Q95 thresholds. The held-out trajectories vary substantially across corruption types, but the warning behavior closely matches the primary analysis.

Across the 15 LOCO folds, there were no early warnings, three warnings at failure, seven late warnings, and three missed warnings. Two corruptions, brightness and saturation, produced neither an observed failure nor a warning. The agreement between the primary analysis and LOCO analysis indicates that the lack of advance warning is not simply a consequence of calibrating the threshold on the same corruption being tested.

G. Signal Redundancy and Multivariate Analysis

The five BDI components were strongly correlated with one another, particularly the prediction-based signals. Confidence and margin degradation had a Pearson correlation of 0.9955, while JS divergence and prediction flip rate had a correlation of 0.9994. The correlations between Grad-CAM instability and

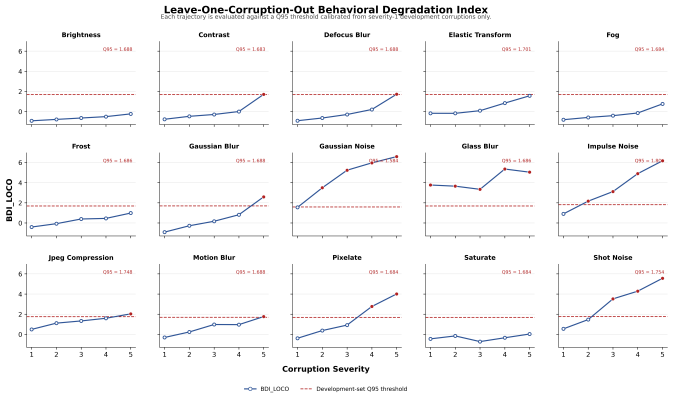


Fig. 7. Leave-One-Corruption-Out BDI trajectories. For each held-out corruption, the BDI is standardized using severity-1 development corruptions only, and the dashed horizontal line denotes the corresponding development-set Q95 threshold.

TABLE V
LEAVE-ONE-CORRUPTION-OUT WARNING OUTCOMES.

Held-out corruption	Failure severity	First BDI warning	Lead time	Outcome
Brightness	–	–	–	No failure / no warning
Contrast	5	5	0	Warning at failure
Defocus blur	5	5	0	Warning at failure
Elastic transform	4	–	–	Missed warning
Fog	5	–	–	Missed warning
Frost	4	–	–	Missed warning
Gaussian blur	4	5	-1	Late warning
Gaussian noise	1	2	-1	Late warning
Glass blur	1	1	0	Warning at failure
Impulse noise	1	2	-1	Late warning
JPEG compression	2	5	-3	Late warning
Motion blur	3	5	-2	Late warning
Pixelation	3	4	-1	Late warning
Saturation	–	–	–	No failure / no warning
Shot noise	2	3	-1	Late warning

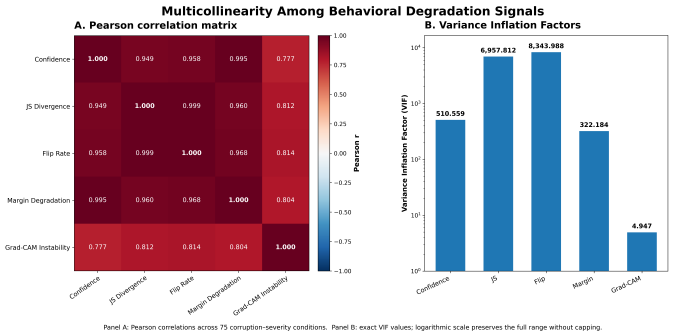


Fig. 8. Multicollinearity among the five BDI component signals. (A) Pearson correlation matrix across the 75 corruption–severity conditions. (B) Variance Inflation Factors shown on a logarithmic scale.

the prediction-based signals were lower, ranging from 0.7766 to 0.8144.

The corresponding VIF analysis confirmed substantial multicollinearity. The largest VIFs were observed for flip rate (8343.988) and JS divergence (6957.812), followed by confidence (510.559) and margin degradation (322.184). Grad-CAM had a substantially lower, although still non-negligible, VIF of 4.947. These results indicate that the prediction-based signals contain considerable redundant information, whereas Grad-CAM contributes a comparatively distinct dimension.

A five-signal ordinary least-squares regression was then fit-

TABLE VI
DESCRIPTIVE FIVE-SIGNAL REGRESSION AND LEAVE-ONE-SIGNAL-OUT REGRESSION RESULTS. COEFFICIENTS ARE REPORTED WITHOUT INFERENTIAL STANDARD ERRORS OR p VALUES BECAUSE THE 75 CONDITION-LEVEL OBSERVATIONS ARE CLUSTERED WITHIN 15 CORRUPTION TYPES AND THE PREDICTORS ARE HIGHLY COLLINEAR.

Full five-signal regression			
Signal	Coefficient		
Intercept	0.0010		
Confidence	0.1323		
JS	0.2883		
Flip	0.8035		
Margin	-0.0554		
Grad-CAM	-0.1265		
$R^2 = 0.999951$, adjusted $R^2 = 0.999947$			
Leave-one-signal-out regression			
Removed signal	R^2	Adjusted R^2	ΔR^2
Grad-CAM	0.999841	0.999832	0.000109
Flip	0.999870	0.999863	0.000080
JS	0.999946	0.999943	0.000005
Confidence	0.999946	0.999943	0.000005
Margin	0.999948	0.999945	0.000003

ted as a descriptive condition-level analysis of accuracy degradation. The resulting model explained almost all observed variation, with $R^2 = 0.999951$ and adjusted $R^2 = 0.999947$. The estimated coefficients were 0.1323 for confidence, 0.2883 for JS divergence, 0.8035 for flip rate, -0.0554 for margin degradation, and -0.1265 for Grad-CAM instability.

However, the 75 observations are clustered within 15 corruption types, and the VIF values demonstrate severe multicollinearity. Therefore, the coefficients are not interpreted as independent predictive or causal effects. The regression is used only to characterize the in-sample association and redundancy among the five condition-level signals.

H. Leave-One-Signal-Out Ablation

The leave-one-signal-out analysis provides a complementary assessment of component contribution. Removing Grad-CAM produced the largest decrease in the BDI–severity association, changing Spearman correlation from 0.5239 to 0.5200 ($\Delta\rho = -0.0039$). Removing flip rate produced the second-largest decrease ($\Delta\rho = -0.0009$). In contrast, removing JS divergence, confidence, or margin degradation slightly increased the Spearman association by 0.0009, 0.0009, and 0.0013, respectively.

The regression ablation showed the same qualitative pattern. Removing Grad-CAM produced the largest reduction in R^2, from 0.999951 to 0.999841, followed by removal of flip rate, which reduced R^2 to 0.999870. Removing JS, confidence, or margin produced only very small changes in R^2.

Taken together, these results suggest that the prediction-based components contain substantial overlapping information, while Grad-CAM provides the most distinctive contribution among the five signals. Nevertheless, the effect sizes of the ablations are small because the components collectively explain highly redundant variation in the condition-level degradation measurements.

I. Summary of Findings

The results provide four principal findings. First, corruption produces measurable instability in model explanations, with Grad-CAM instability varying systematically across corruption types and severity levels. Second, the proposed BDI shows a positive descriptive association with corruption severity, with positive severity associations observed for all 15 corruptions and a mean corruption-wise monotonicity of 0.9000.

Third, the proposed multilevel degradation hierarchy is not universal. Only 20.0% of corruptions follow the complete $L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4$ sequence, whereas 53.3% exhibit reversal or non-monotonic progression. This demonstrates that different corruption mechanisms can produce qualitatively different degradation pathways.

Finally, although BDI is strongly associated with increasing corruption severity, the primary Q95 warning mechanism does not provide advance warning of the defined performance failure. No early warnings were observed for BDI or any other monitored signal, and the same pattern persisted under leave-one-corruption-out calibration. The combined correlation, VIF, regression, and ablation analyses further show that the prediction-based signals are highly redundant, while Grad-CAM provides the most distinct contribution to the composite degradation representation.

V. DISCUSSION

A. Interpretation of Results

The experimental results demonstrate that the proposed Behavioral Degradation Index (BDI) provides a compact measure of behavioral instability under image corruption. The BDI combines five complementary signals, namely confidence change, Jensen–Shannon divergence, prediction flip rate, prediction-margin degradation, and Grad-CAM explanation instability. Each signal is standardized across the 75 corruption–severity conditions and combined using fixed equal weights, providing a transparent aggregation mechanism.

The mean BDI increased consistently across the five severity levels, from -0.6517 at severity 1 to 0.7504 at severity 5. The aggregate Spearman correlation between BDI and corruption severity was 0.5239, indicating a positive monotonic association between increasing corruption severity and overall behavioral degradation. This result supports the use of BDI as a severity-sensitive measure of model behavior under corruption.

However, the multilevel analysis also shows that degradation does not necessarily follow a single deterministic progression for every corruption type. Different corruptions produced different transitions across degradation levels. Therefore, the BDI should be interpreted as an aggregate characterization of behavioral degradation rather than as a universal indicator that follows the same progression for all corruption mechanisms.

The adaptive-threshold analysis provides an additional perspective. Although the BDI and the other individual signals

were able to identify degraded behavior at or after performance failure for several corruptions, none of the evaluated signals produced an early warning under the primary Q_{95} threshold formulation. The LOCO analysis further demonstrated that warning behavior varied substantially across corruption types. These findings indicate that the current framework is more reliable for characterizing and comparing degradation than for providing consistent advance prediction of future accuracy failure.

The regression analysis produced a very high in-sample coefficient of determination ($R^2 = 0.999951$). The leave-one-signal-out analysis also resulted in only modest changes in model fit. Because the observations are clustered by corruption type and the predictors are highly collinear, these regression results are interpreted as evidence of strong condition-level association and signal redundancy rather than as evidence of independent predictive power or causal influence on model failure.

B. Implications

The results suggest that behavioral degradation can be characterized using multiple complementary observations rather than relying only on classification accuracy. Prediction confidence, distributional change, prediction instability, margin degradation, and explanation instability capture different aspects of how a classifier responds to corrupted inputs.

The proposed BDI therefore provides a unified representation that can be used to compare degradation across corruption types and severity levels. Its fixed equal-weight construction also makes the index straightforward to reproduce and interpret. Nevertheless, the observed variation across corruption pathways indicates that BDI should currently be regarded primarily as a diagnostic and comparative measure rather than a universal early-warning mechanism for model failure.

C. Limitations

Several limitations should be considered when interpreting the results. First, the experiments are conducted using CIFAR-10 and CIFAR-10-C with a single ResNet-50 model. Consequently, the observed behavior may not generalize directly to other datasets, architectures, or application domains.

Second, the BDI uses fixed equal weights for its five component signals. This choice provides transparency and avoids learning the weights from the evaluation data, but it does not establish that equal weighting is optimal for every model or task.

Third, the threshold-based warning analysis depends on the selected percentile calibration strategy and the corruption-specific experimental setting. In particular, the primary Q95 thresholds are estimated from only 15 severity-1 condition-level observations, while each LOCO threshold is estimated from only 14 development observations. The resulting warning rates should therefore be interpreted as small-sample empirical evidence rather than precise estimates of deployment-level detection performance. The absence of consistent early warnings under the evaluated thresholds does not imply that behavioral signals can never provide predictive information; rather, it shows that the current threshold formulation did not reliably anticipate accuracy failure in the evaluated experiments.

Finally, the Grad-CAM component depends on the selected explanation method and its preprocessing and correspondence procedure. Different explanation techniques may measure explanation instability differently and could consequently affect the resulting BDI. The clean–CIFAR-10-C correspondence is also recovered algorithmically rather than supplied directly as metadata, so its matching procedure is an additional source of methodological dependence.

D. Future Work

Future work should first evaluate the BDI across a broader range of datasets, architectures, and distribution-shift scenarios. Testing models beyond the ResNet-50/CIFAR-10 setting would help determine whether the observed relationship between behavioral degradation and corruption severity generalizes across model families and task domains.

A second direction is to investigate adaptive or learned weighting strategies for the five BDI components. Instead of using fixed equal weights, future studies could examine task-specific or uncertainty-aware weighting schemes using independent validation data to determine whether particular signals provide greater value for specific degradation scenarios.

Another important direction is to investigate genuine temporal or distribution-shift settings in which behavioral changes can be observed before a future performance failure occurs. Such experiments would provide a stronger test of whether the BDI can serve as a predictive early-warning indicator rather than only a descriptive degradation measure.

Future research can also incorporate uncertainty estimation and probabilistic risk modeling into the warning mechanism. This could allow the framework to express uncertainty in degradation estimates instead of relying only on deterministic percentile thresholds.

Finally, the explanation component can be extended beyond Grad-CAM by evaluating multiple explanation techniques and investigating whether explanation instability provides complementary information to the prediction-based signals. Cross-corruption validation without corruption-specific calibration would also provide a stronger test of the generalizability and practical usefulness of the proposed framework.

VI. Conclusion

This study presented a Behavioral Degradation Index (BDI) for characterizing the behavioral changes of an image classification model under corrupted inputs. The proposed index combines five signals: confidence change, Jensen–Shannon divergence, prediction flip rate, prediction-margin degradation, and Grad-CAM explanation instability. After standardization across 75 CIFAR-10-C corruption–severity conditions, the signals are combined using fixed equal weights to produce a single degradation measure.

The experimental results show that the BDI increases with corruption severity, with the mean BDI changing from

-0.6517 at severity 1 to 0.7504 at severity 5. The aggregate Spearman correlation of 0.5239 further demonstrates a positive association between BDI and corruption severity. At the same time, the multilevel analysis shows that different corruption types can follow different degradation pathways, highlighting the heterogeneous nature of model behavior under distribution shift.

The threshold and LOCO analyses provide an important qualification to these findings. While behavioral signals can identify degradation at or around performance failure for several corruption types, the evaluated Q_{95} formulation did not provide consistent early warnings before failure. Thus, the current results support the BDI as a diagnostic and comparative measure of behavioral degradation rather than as a universally reliable predictor of future accuracy failure.

Overall, the proposed framework provides a reproducible way to combine prediction-based and explanation-based behavioral signals into a unified degradation index. The findings demonstrate the potential of behavioral analysis to complement conventional accuracy-based evaluation while also emphasizing the need for broader validation, adaptive weighting, improved uncertainty modeling, and realistic future-failure prediction. Future extensions across architectures, datasets, and distribution-shift settings can further establish the generality and practical value of the BDI framework.

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